

The Price of Local Ordering in Two-Stage Assembly Flow Shops

Eric Bowman^{1*}

¹King, Berlin, Germany.

Corresponding author(s). E-mail(s): eric.bowman@king.com;

Abstract

In the divisible-work two-stage assembly flow shop, permutation schedules—one job order shared by every machine—dominate pointwise, so adopting a shared order is never worse. This paper quantifies the converse: how much can abandoning the shared order cost? In the fluid homogeneous model we first pin the optimum exactly: every schedule satisfies $k C_{\max} \geq W + w_{\max}$, and the synchronized schedule attains it under *every* job order; consequently, under arbitrarily distributed random work, a fixed shared order—chosen with no foresight—stochastically dominates every scheduling policy on makespan, and the permutation class is stochastically complete for all regular objectives. Against this exact baseline the price of local ordering is sharp: arbitrary per-machine orders combined with a downstream stage that may hold capacity idle for a preferred job cost exactly $2W/(W + w_{\max}) \rightarrow 2$ on makespan, while *any* work-conserving downstream stage—regardless of its processing order—caps the price at $2 - 1/m$, where m is the number of stage-1 machines; both constants are tight. The same dichotomy governs total flowtime, with tight constants 3 and $3 - 2/m$; for weighted completion time no constant-factor ceiling exists. The structural conclusion is that the expensive downstream behavior is *work-withholding*, not misordering: a stage that idles while released work waits forfeits as much again as all upstream misalignment combined. Every constant is verified by exhaustive enumeration over the instance family of the companion paper, where the worst cases match the theory exactly, and the two bracketing results are formalized in Lean 4.

Keywords: assembly flow shop, permutation schedule, divisible work, price of anarchy, work conservation, stochastic scheduling

MSC Classification: 90B35

1 Introduction

Dominance theorems are one-sided. For the two-stage assembly flow shop with divisible work— m parallel stage-1 machines of k identical workers feeding a single assembly stage of k workers—for every schedule there is a *permutation schedule*, with one job order shared by all stage-1 machines, that completes every job at the assembly stage no later (Bowman, 2026a). A companion paper maps where such dominance survives on deeper stage structures (Bowman, 2026b). These results say that coordinating on a shared order is *never worse*. They are silent on the question any adopter of the policy actually asks: how much worse is it *not* to coordinate?

This paper answers that question exactly, in the fluid (real-divisible) homogeneous form of the model. Three groups of results.

First, the baseline is the true optimum (Section 3). Every schedule—any per-machine orders, any work divisions, any downstream discipline, preemptive or not, idle or not—satisfies

$$k C_{\max} \geq W + w_{\max},$$

where W is the total work and $w_{\max}$ the largest job, and the synchronized schedule attains this bound under *every* shared order π (Theorem 2). Two consequences follow. Under arbitrarily distributed random work—any joint law—a fixed shared order, chosen before any work is known, is pathwise makespan-optimal, hence stochastically dominant over every policy including clairvoyant ones: for the date the whole portfolio of jobs lands, foresight is worth nothing and the choice of shared order is worth nothing; only *sharedness* matters (Corollary 4). And the pointwise dominance theorem of Bowman (2026a) couples, realization by realization, into a stochastic completeness statement: no policy beats the permutation class on any regular objective in any stochastic sense (Theorem 5). All prices below are therefore measured against the exact optimum, not against a possibly suboptimal synchronized baseline.

Second, the price of local ordering on makespan is an exact dichotomy (Section 4). If every stage-1 machine chooses its own order and the downstream stage may *withhold* work—idle while released work waits, as a stage does when it insists on processing in a preferred order whose next job has not yet arrived—the worst-case makespan ratio is exactly $2W/(W + w_{\max})$, approaching 2 (Theorem 7). If instead the downstream stage is *work-conserving*—never idle while released work is unfinished, but otherwise free to process in any order whatsoever—the price drops to $2 - 1/m$, and this is tight even against a clairvoyant downstream (Theorem 9). The proof of the upper bound never uses the downstream *order*; only non-idleness. The structural conclusion, which we believe is the practically important one, is that for makespan the expensive downstream sin is withholding, not misordering; a downstream stage that holds capacity idle for its preferred job forfeits as much again as all upstream misalignment combined.

Third, the same dichotomy governs flowtime, and no ceiling exists for weighted completion time (Sections 5–6). For equal works the total-flowtime price is exactly $(3n + 1)/(n + 3) \rightarrow 3$ under full anarchy and $3 - 2/m$ under a work-conserving downstream, both tight. For weighted completion time the price is unbounded: concentrating value on one small job and ordering it locally multiplies the weighted objective by $\Theta(W/w_{\max})$.

Every constant above is exact at finite sizes in a way that can be checked by brute force, and we have checked it: over the 39-configuration exhaustive enumeration family of Bowman (2026a) (roughly 4.0×10^6 schedules), with the unsynchronized population split by downstream discipline, the worst cases agree with the theory to the digit, including two structural predictions — the two downstream classes must coincide once $m \geq n$, and the withholding class must be empty at $n = 2$ (Section 7). The two bracketing results (Theorem 2’s lower bound, and the universal upper bound behind Theorem 7) are additionally formalized in Lean 4 with no unproven steps.

1.1 Related work

The question belongs to the price-of-anarchy tradition of Koutsoupias and Papadimitriou (1999): compare a coordinated optimum against the worst outcome of uncoordinated choices. Our setting differs from game-theoretic scheduling in that we do not model incentives or equilibria; “anarchy” here is simply the absence of a shared order, which is how the phenomenon presents in the motivating applications (portfolios of cross-team initiatives; multi-stage processing pipelines). The two-stage assembly flow shop originates with Lee et al. (1993) and Potts et al. (1995), who proved permutation dominance and approximation results in the unit-capacity case; see Hariri and Potts (1997) for exact methods and Bowman (2026a) for the divisible-work generalization and the literature between.

The constant $2 - 1/m$ of Theorem 9 has the same form as Graham’s bound for list scheduling on m identical machines (Graham, 1966; Graham et al., 1979), and the resemblance is not coincidental: both results bound the damage of arbitrary orderings by a busy-period argument in which work conservation is the only property used. The settings are different—Graham’s m counts parallel machines executing the jobs, our m counts upstream machines feeding an assembly stage, and our ratio is measured against an exactly known optimum—but the mechanism, that non-idleness alone buys a factor- $(2 - 1/m)$ guarantee, appears to be the same phenomenon in two guises. The stochastic results connect to the classical observation that simple index policies retain optimality under uncertainty (Rothkopf, 1966; Smith, 1956; Pinedo, 2016); Corollary 4 is an extreme instance in which the optimal policy requires no distributional information at all. Divisible (malleable) work is surveyed by Drozdowski (2009).

2 Model and policy classes

There are n jobs; job j requires $w_j > 0$ units of work on each of m stage-1 machines and w_j units on a single stage-2 (assembly) machine. Every machine has k identical unit-rate workers. Work is divisible: a machine that devotes all k workers to one job processes it at rate k . Write $W = \sum_j w_j$ and $w_{\max} = \max_j w_j$. The *assembly constraint*: job j may not be processed at stage 2 before all m stage-1 machines have completed it; the time this happens is j ’s *availability* a_j . All jobs are present at time zero. This is the homogeneous fluid form of the model of Bowman (2026a); the integer-divisible form used there for the pointwise theorem appears here only in Theorem 5 and in the computational section.

A schedule is *team-sequential* if every machine processes one job at a time at full rate k , stage-1 machines without inserted idleness. (Stage-1 idleness and unbalanced work divisions can only increase availabilities and are briefly discussed after each upper bound; the lower bound of Theorem 2 holds with no restriction at all.) Within team-sequential schedules we distinguish the downstream discipline:

Definition 1 (Policy classes) A team-sequential schedule is:

- *synchronized* if all stage-1 machines and the stage-2 machine process the jobs in one common order π ;
- *work-conserving downstream* if the stage-2 machine is never idle while some released job is unfinished—its processing order is otherwise arbitrary;
- *work-withholding downstream* if it processes the jobs back-to-back in some fixed preferred order, each job starting as soon as the stage is free *and* the job is available—so the stage idles exactly when its order’s next job has not yet arrived, even while other released work waits. (This is the behavior of a stage that enforces a priority list internally; it is the only inserted idleness we allow downstream, and the enumeration model of Section 7 matches it.)

Upstream anarchy means the stage-1 machines use arbitrary, possibly different orders. *Full anarchy* means upstream anarchy with a work-withholding downstream.

Objectives are functions of the stage-2 completion vector $(C_j)_j$: makespan $C_{\max} = \max_j C_j$, total flowtime $\sum_j C_j$, and weighted completion time $\sum_j v_j C_j$ for non-negative values v_j .

3 The exact optimum, deterministically and stochastically

Theorem 2 (The optimum, and its order-independence)

- Every schedule—team-sequential or not, with arbitrary work divisions, inserted idleness, and downstream discipline, preemptive or not—satisfies $k C_{\max} \geq W + w_{\max}$.
- For every order π , the synchronized team-sequential schedule with order π has $C_{\max} = (W + w_{\max})/k$.

Hence the optimal makespan is exactly $(W + w_{\max})/k$, attained by the shared order—any shared order.

Proof (a) Let M be a job of maximal work and let $a = a_M$ be its availability; fix one stage-1 machine that completes M at time a (if several, any). By time a that machine has processed at most ak units of work, of which $w_{\max}$ belong to M . A job is eligible for stage 2 only when complete on *all* machines, in particular on this one, so the total work of jobs other than M available by time a is at most $ak - w_{\max}$, and stage 2 can have completed at most that much

by time a . At least $W - (ak - w_{\max})$ units of stage-2 work remain at time a , processed at rate at most k :

$$C_{\max} \geq a + \frac{W - ak + w_{\max}}{k} = \frac{W + w_{\max}}{k},$$

independent of a . The argument uses only capacity counting; it is indifferent to work divisions, idleness, preemption, and the number of machines.

(b) Under the shared order π , stage 1 completes the job at position s at time S_s/k where $S_s = \sum_{t \leq s} w_{\pi(t)}$. Writing F_s for its stage-2 completion and $M_s = \max_{t \leq s} w_{\pi(t)}$, induction on s gives $F_s = (S_s + M_s)/k$: indeed $F_s = \max(S_s, S_{s-1} + M_{s-1})/k + w_{\pi(s)}/k = (S_{s-1} + \max(M_{s-1}, w_{\pi(s)}))/k + w_{\pi(s)}/k = (S_s + M_s)/k$. At $s = n$ this is $(W + w_{\max})/k$ for every π . $\square$

Remark 3 (Lean formalization) Part (a) is machine-checked in Lean 4 over the concrete integer-divisible machine model of Bowman (2026a), with stage 2 abstracted by a cumulative-progress structure whose axioms (progress capped by the work, zero before release, total rate at most k) are satisfied by every physical execution; the formal statement, $kT \geq W + w_M$ for every job M , is correspondingly stronger than (a). The formalization also revealed that the choice of machine in the proof is immaterial: any machine bounds the released work, because every completion lies below the availability. The development has no unproven steps and uses only the standard axioms.

The optimum being attained by every shared order has a stochastic consequence that we state in full generality.

Corollary 4 (A fixed shared order is stochastically optimal for makespan) *Let the works $(W_1, \dots, W_n)$ be random with an arbitrary joint distribution, and fix any order π in advance. Then, pathwise, the synchronized schedule with order π achieves $C_{\max} = (W + W_{\max})/k \leq C_{\max}(S)$ for every scheduling policy S , including policies with full knowledge of the realization. In particular the fixed shared order stochastically dominates every policy on makespan.*

Proof Apply Theorem 2 realization by realization: (b) gives the value of the fixed-order schedule, (a) bounds every competitor below by the same value. $\square$

For objectives beyond makespan a single fixed order no longer suffices, but the class of shared-order schedules remains unbeatable, again under arbitrary randomness. The following statement is in the integer-divisible model of Bowman (2026a), whose pointwise theorem it couples; it is the formal sense in which all prices in this paper are measured against an optimum that stochastic work cannot dislodge.

Theorem 5 (Stochastic completeness of the permutation class) *Let the works have an arbitrary joint distribution and let S be any scheduling policy, possibly randomized and clairvoyant. There is a works-measurable permutation-schedule policy S^* with $C_j(S^*) \leq C_j(S)$ for every job j , almost surely. Consequently, for every coordinatewise-monotone objective f , $f(C(S^*)) \leq f(C(S))$ almost surely, and no policy beats the permutation class in expectation, in any quantile, or in the usual stochastic order, for any regular objective.*

Proof The construction of Bowman (2026a) maps each realization of the works and of S 's schedule to a permutation schedule dominating it pointwise; the permutation used is obtained by sorting the realized availabilities, so the map is measurable (there are finitely many orders, and completion times are measurable in the works). Composing S with this map yields S^* ; the domination is pointwise on the same probability space, i.e. a coupling, and all claimed comparisons follow. $\square$

Remark 6 Theorem 5 is existential in the order, which may depend on the realization. Corollary 4 shows that for makespan the dependence vanishes. Whether a fixed *value-ordered* permutation is optimal in expectation for weighted completion time under exchangeable works—the natural analogue of weighted-shortest-expected-processing-time results (Rothkopf, 1966)—is open; see Section 8.

4 The price of local ordering: makespan

Theorem 7 (Full anarchy doubles the schedule, exactly) *Over team-sequential schedules with upstream anarchy and a work-withholding downstream,*

$$\sup_{\text{instances, schedules}} \frac{C_{\max}}{(W + w_{\max})/k} = \sup_{\text{instances}} \frac{2W}{W + w_{\max}} = 2,$$

with the inner supremum attained for every instance with $m \geq 2$: no team-sequential schedule exceeds $2W/k$, and for every instance some unsynchronized schedule reaches it.

Proof Upper bound. Stage-1 machines are work-conserving and process at rate k , so every machine finishes all its work at W/k ; hence $a_j \leq W/k$ for every j . A downstream stage that processes its jobs back-to-back in any order σ , waiting only for availabilities, satisfies the max-plus recursion $F_{\sigma(s)} = \max(F_{\sigma(s-1)}, a_{\sigma(s)}) + w_{\sigma(s)}/k$, whence by induction $C_{\max} \leq \max_j a_j + W/k \leq 2W/k$.

Lower bound. Fix any job J . Let stage-1 machine 1 process J last (so $a_J = W/k$) and machine 2 use any different order (so the schedule is unsynchronized). Let the downstream stage process J *first*: it idles until W/k , then has all W units to do with no further idleness, finishing at exactly $2W/k$. Against the optimum $(W + w_{\max})/k$ of Theorem 2 the ratio is $2W/(W + w_{\max})$, which tends to 2 as $w_{\max}/W \rightarrow 0$. $\square$

Remark 8 The upper bound is also machine-checked in Lean 4 at the release-time interface (makespan at most the largest availability plus the total downstream work, for every downstream order), along with the availability bound and computed witnesses for the smallest anti-aligned instance. With unbalanced work divisions the *lower* bound only improves for the adversary; the sweep of Section 7 shows split badness stacking on top of the ordering price exactly where divisions are not forced balanced.

The construction above needs the downstream stage to *wait* for J . The next theorem shows this is the only way to get anywhere near a factor of two.

Theorem 9 (Work conservation caps the price at $2 - 1/m$) *Over team-sequential schedules with upstream anarchy and ANY work-conserving downstream discipline,*

$$C_{\max} \leq \left(2 - \frac{1}{m}\right) \frac{W}{k} + \frac{w_{\max}}{k},$$

and the constant is tight: for equal works w and $m \mid n$ there are stage-1 orders under which every downstream discipline—work-conserving, withholding, or clairvoyant—has $C_{\max} \geq (2 - 1/m)nw/k + w/k$. Hence the price of upstream anarchy under a work-conserving downstream is exactly $2 - 1/m$ asymptotically.

Proof Upper bound. Let t_1 be the last instant stage 2 is idle before $C_{\max}$ (if stage 2's work is complete at its last idle instant, then $C_{\max}$ is at most the last availability, at most W/k , and we are done). At t_1 , work conservation means every available job is already finished at stage 2, so the remaining stage-2 work is at most $W - w(A(t_1))$, where $A(t)$ is the set of jobs available by t and $w(\cdot)$ sums their works; stage 2 is busy at rate k on $[t_1, C_{\max}]$, giving $C_{\max} \leq t_1 + (W - w(A(t_1)))/k$.

Since stage 2 idles at t_1 with some unfinished job not yet available, some stage-1 machine is still busy; stage-1 machines have no upstream and are work-conserving, so that machine has processed exactly $t_1 k \leq W$. Every still-busy machine i has processed exactly $t_1 k$ units, of which at most $w_{\max}$ sit in its single in-progress job, so its set $F_i(t_1)$ of finished jobs satisfies $w(F_i(t_1)) \geq t_1 k - w_{\max}$ (machines already done satisfy this trivially). A job is available iff finished on all m machines, so by inclusion–exclusion for the work measure,

$$w(A(t_1)) = w\left(\bigcap_i F_i(t_1)\right) \geq \sum_i w(F_i(t_1)) - (m-1)W \geq m(t_1 k - w_{\max}) - (m-1)W.$$

Set $T^* = (m-1)W/(mk) + w_{\max}/k$. If $t_1 \leq T^*$ then $C_{\max} \leq t_1 + W/k \leq (2 - 1/m)W/k + w_{\max}/k$. If $t_1 \geq T^*$ then $C_{\max} \leq t_1(1 - m) + (mW + mw_{\max})/k$, which is decreasing in t_1 and equals $(2 - 1/m)W/k + w_{\max}/k$ at $t_1 = T^*$.

Lower bound (rotation construction). Equal works w , $m \mid n$. Let machine i process the jobs in the cyclic rotation of $(1, \dots, n)$ by $(i-1)n/m$ positions. The m rotation offsets are evenly spaced, so job j 's positions across the machines form the set $\{(j \bmod (n/m)) + r n/m + 1 : r = 0, \dots, m-1\}$; its *worst* position is at least $(m-1)n/m + 1$. Hence no job is available before $T_0 = ((m-1)n/m + 1)w/k$, and stage 2—however scheduled—must process all $W = nw$ units after T_0 : $C_{\max} \geq T_0 + nw/k = (2 - 1/m)nw/k + w/k$. Against the optimum $(n+1)w/k$ the ratio tends to $2 - 1/m$. (For $m \nmid n$, rotating by $\lfloor n/m \rfloor$ degrades the constant by $O(w/k)$.) $\square$

Remark 10 (The dichotomy) The upper bound of Theorem 9 never refers to the downstream order; non-idleness is the only property used. So for makespan the downstream behaviors split into exactly two classes: any work-conserving stage, even one that ignores priorities entirely, is protected at $2 - 1/m$; only a work-withholding stage—one that holds capacity idle for its preferred job—reaches the full price of Theorem 7. With two upstream machines the numbers are stark: upstream misalignment alone costs at most 50%; downstream withholding doubles it. In operational terms, enforcing a priority order *inside* a downstream stage is precisely a withholding discipline whenever the preferred job has not yet arrived, and it forfeits as much again as all upstream misalignment combined. The resemblance of $2 - 1/m$ to Graham's list-scheduling bound (Graham, 1966) is discussed in Section 1; both are busy-period consequences of work conservation alone.

5 The price of local ordering: flowtime

For flowtime we take equal works w (so $W = nw$) and a *non-preemptive* one-job-at-a-time downstream stage; the spacing step below fails under preemptive interleaving, whose constant we conjecture is the same (Section 8). The synchronized flowtime is $\sum_s (s+1)w/k = n(n+3)w/(2k)$ by Theorem 2(b), and it is optimal for equal works: the earliest stage-2 completion is at least $2w/k$ (the first job needs w/k at each stage), and the non-preemptive downstream completes jobs at least w/k apart, so every schedule has $C_{(s)} \geq (s+1)w/k$.

Theorem 11 (Full anarchy triples flowtime, exactly) *Over team-sequential schedules with a non-preemptive downstream, for equal works,*

$$\sup \frac{\sum_j C_j}{n(n+3)w/(2k)} = \frac{3n+1}{n+3} \rightarrow 3,$$

the supremum over upstream anarchy with a withholding downstream, attained for every n when $m \geq 2$.

Proof Upper bound. Stage 2 processes one job at a time, each taking w/k , so the sorted completions satisfy $C_{(s+1)} \geq C_{(s)} + w/k$; with $C_{(n)} = C_{\max} \leq 2nw/k$ (Theorem 7), $C_{(s)} \leq (n+s)w/k$ and $\sum_j C_j \leq (3n^2 + n)w/(2k)$ for *every* team-sequential schedule. The ratio to $n(n+3)w/(2k)$ is $(3n+1)/(n+3)$.

Lower bound. In the construction of Theorem 7's lower bound, stage 2 idles until nw/k and then completes one job per w/k : completions are exactly $(n+s)w/k$ for $s = 1, \dots, n$, meeting the upper bound term by term. $\square$

Theorem 12 (Work conservation caps the flowtime price at $3 - 2/m$) *Under the hypotheses of Theorem 11 but with any work-conserving downstream, for equal works and $m \mid n$,*

$$\sum_j C_j \leq \left(\frac{3}{2} - \frac{1}{m}\right) \frac{n^2 w}{k} + O(n),$$

and the rotation construction attains $\sum_j C_j = (\frac{3}{2} - \frac{1}{m})n^2 w/k + O(n)$ under every downstream discipline. The price is exactly $3 - 2/m$ asymptotically.

Proof Upper bound. By Theorem 9, $C_{(n)} \leq ((2 - 1/m)n + 1)w/k$; spacing gives $C_{(s)} \leq ((1 - 1/m)n + s + 1)w/k$; summing yields the claim.

Lower bound. In the rotation instance the m jobs with residue c become available exactly at slot $(m-1)n/m + c + 1$, so m new jobs arrive per slot from T_0 on while stage 2 completes one per slot: it never starves after T_0 , and completions are exactly $T_0 + sw/k$, summing to $nT_0 + n(n+1)w/(2k) = (\frac{3}{2} - \frac{1}{m})n^2 w/k + O(n)$. Since no job is available before T_0 , no downstream discipline does better than these completions term by term. $\square$

The pattern of Remark 10 thus persists in the flowtime currency— $3 - 2/m$ for any work-conserving downstream versus 3 for a withholding one—with the same interpretation: withholding forfeits as much again as all upstream misalignment.

6 No ceiling for weighted completion time

Proposition 13 (The weighted price is unbounded) *For every B there is an instance and an unsynchronized team-sequential schedule whose weighted completion time exceeds B times the optimum.*

Proof Put value v on a single job J of work w and negligible value on the others. The synchronized schedule with J first completes it at $2w/k$, so the optimum is at most $2vw/k + o(v)$. The construction of Theorem 7 applied to J completes it at $2W/k$, giving weighted objective at least $2vW/k$. The ratio is $W/w + o(1)$, unbounded as the number of jobs grows. $\square$

Schedule damage is capped at a factor of two (Theorem 7); value damage has no cap, and it grows exactly with the concentration of value. Combined with Corollary 4 this separates the two roles of the shared order: *any* shared order is optimal for the date all work lands; a *correct* (value-aware) shared order is what protects when value lands.

7 Computational verification

The constants above are exact at finite sizes, so they can be confronted with exhaustive enumeration. We use the 39-configuration instance family of Bowman (2026a) ($n \leq 5$ jobs, $m \in \{2, 3\}$ stage-1 machines, $k \in \{2, 3\}$ workers, symmetric, asymmetric, value-weighted and machine-heterogeneous works; about 4.0×10^6 schedules in total), with every combined schedule classified as synchronized or unsynchronized and, within the unsynchronized population, the downstream order classified as availability-consistent (work-conserving in the integer model) or re-ordered (withholding). The integer model forces balanced divisions exactly when $w_j = k$; in those configurations the fluid constants must be—and are—attained exactly.

Table 1 Worst-case makespan price (per cent over the optimum) of the unsynchronized population, split by downstream discipline, on the division-forced configurations, against the finite- n theory of Theorems 7 and 9.

Configuration	Work-conserving		Withholding	
	observed	theory	observed	theory
$n = 3, m = 2$	+25.0%	5/4	+50.0%	6/4
$n = 4, m = 2$	+40.0%	7/5	+60.0%	8/5
$n = 5, m = 2$	+33.3%	8/6	+66.7%	10/6
$n = 3, m = 3$	+50.0%	coincide	+50.0%	6/4
$n = 4, m = 3$	+40.0%	7/5	+60.0%	8/5
$n = 2, \text{any } m$	+33.3%	4/3	class empty (forced)	

Beyond the exact agreement of every entry, two structural predictions are reproduced: at $n = 3, m = 3$ the two downstream classes coincide, as they must once $m \geq n$

(the rotation construction then defeats every downstream discipline); and at $n = 2$ the withholding class is empty, as it must be (availabilities tie, so every downstream order is availability-consistent). The flowtime worst cases similarly match Theorems 11 and 12 exactly where divisions are forced ($+100.0\% = (3n + 1)/(n + 3)$ at $n = 5$; $+50.0\%$ for the work-conserving class, the rotation value). Configurations with division freedom exceed the balanced-split constants in both classes, quantifying how division badness stacks on top of ordering badness.

Averages tell the operational story: across the full family, the mean makespan price of an unsynchronized schedule is 39.6% (per-configuration means 20.0–63.2%), the mean flowtime price 60.6%, and the mean weighted price 69.2% (worst +475.5%, under value skew 100:1:1); on average across configurations, more than 99.5% of unsynchronized schedules are strictly later than the optimum on every objective (no configuration falls below 95.6%). Within the unsynchronized population, conditioning on an availability-consistent downstream roughly halves the mean price (e.g. from +53.1% to +26.3% at $n = 5$, $m = 2$)—work conservation alone recovers about half the damage of upstream misalignment, before any upstream coordination.

The two bracketing results are formalized in Lean 4 over the concrete machine model of Bowman (2026a): the universal lower bound $k C_{\max} \geq W + w_M$ for every job M (Theorem 2(a)), with stage 2 abstracted so that any physical execution satisfies the axioms; and the universal upper bound at the release-time interface behind Theorem 7, together with computed witnesses for the smallest anti-aligned instance ($2W$ versus $W + w_{\max}$). Both developments contain no unproven steps. Code, enumeration outputs, and the Lean development are in the archived artifact (see Declarations).

8 Discussion

Together with Bowman (2026a,b), the results complete a two-sided account of the shared order in this model. The dominance theorem says adopting it is free; this paper says declining it costs 40% on average in the exhaustive family, at most a factor of two on the schedule (exactly), without limit on concentrated value—and that the two halves of the price have different owners. Upstream misalignment is worth $2 - 1/m$; everything beyond is the downstream stage’s choice to hold capacity idle for a preferred job. For practice the asymmetry is the message: a downstream stage that simply keeps pulling whatever has arrived is protected by Theorem 9 no matter how badly it orders; the behavior to eliminate is waiting—which is what enforcing a priority list inside a stage operationally means whenever the list’s next item has not arrived. The stochastic results add that none of this depends on knowing the work: a fixed shared order is exactly optimal for the portfolio date under any randomness (Corollary 4), while protecting *value* requires the order to be right (Proposition 13)—sharedness is free insurance, correctness buys value.

Open problems. (i) The flowtime spacing arguments require a non-preemptive downstream; we conjecture the constants 3 and $3 - 2/m$ survive preemption. (ii) The flowtime constants are proved for equal works; the general-works analogues should follow the same pattern. (iii) The expected price under *random* (rather than adversarial) per-machine orders is strictly positive and empirically large (39.6% mean above);

its exact asymptotics are open. (iv) Whether the value-ordered fixed permutation is expectation-optimal for weighted completion time under exchangeable works (the WSEPT analogue) is open. (v) The online version—jobs arriving over time, the shared order maintained by preemptive priority—is the natural continuation and the model closest to the motivating applications.

Declarations

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Author contributions. The sole author conceived the work, developed the proofs, and wrote the manuscript.

Data availability. The enumeration outputs supporting Section 7 are available in the repository below.

Code availability. The simulation code, exhaustive-enumeration suite, and the Lean 4 formalization are available at <https://github.com/ebowman/synchronization-paper-artifact> and archived at <https://doi.org/10.5281/zenodo.20075388>.

Generative AI disclosure. AI-assisted tooling (Anthropic Claude) was used for literature search, drafting support, simulation code, and proof checking; all mathematical content was verified by the author, by exhaustive computational enumeration, and—where stated—by the Lean 4 proof assistant.

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